



Final Year Project Report

AI4VIS: AI-POWERED INTERACTIVE CREATION OF VISUALISATIONS - STANDARD CHARTS

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Abstract

Large Language Models (LLMs) are increasingly being used to support data visualization and analysis through natural language interaction. These models have demonstrated growing capabilities in interpreting, generating, and reasoning over graphical information. However, their reliability and consistency across different visual and chart-related tasks present valuable opportunities for further exploration.

This final-year project evaluates the performance of contemporary LLMs, including ChatGPT, Gemini, Claude, and Mistral, across a range of chart-related tasks. The evaluation focuses on three key areas: visual encoding interpretation, chart-based question answering, and the detection of misleading visualizations. A category-based evaluation framework is developed to systematically assess model responses and identify recurring error patterns.

The study establishes baseline performance benchmarks for each model and provides a comparative analysis of their strengths and limitations. The findings highlight common challenges faced by LLMs in visual reasoning tasks and offer insights into their reliability in data analysis contexts.

Overall, this project contributes to a deeper understanding of the capabilities and limitations of LLMs in visualization-related tasks, supporting their more effective and responsible use in real-world analytical applications.

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Chapter 1

Introduction

1.1 Background

Data visualization is the process of representing data in graphical form to communicate information clearly and effectively. **Standard chart types**, such as bar charts, line charts, and pie charts, are widely used to transform discrete values into visual representations that are easier to interpret and often more intuitive than textual descriptions. When designed appropriately, visualizations can enhance understanding, reveal patterns, and support informed decision-making.

With recent advancements in artificial intelligence, **Large Language Models (LLMs)** have demonstrated significant progress in natural language understanding and generation. These models are increasingly capable of supporting data-related tasks, including generating, interpreting, and reasoning over visual representations through natural language interaction. Such capabilities have the potential to lower the barrier to chart creation, enabling users to generate and analyze visualizations without requiring extensive technical expertise.

Despite their growing adoption, the reliability of LLMs in handling chart-related tasks remains an open research challenge. Tasks such as interpreting visual encoding elements, answering questions based on chart content, and identifying misleading or incorrect visualizations require accurate visual reasoning and contextual understand-

ing. Limitations in these areas may lead to incorrect interpretations, flawed insights, and the potential propagation of misleading information.

Several commercial and open-source LLMs, including **ChatGPT, Gemini, Claude, and Mistral**, are now widely accessible and increasingly used for visualization-related assistance. While existing studies provide initial insights into their capabilities, there remains a lack of comprehensive and systematic evaluation across diverse chart-based tasks.

This final-year project is motivated by the need to better understand the **strengths, limitations, and behavioral patterns of contemporary LLMs** in graphical reasoning contexts. By evaluating model performance across multiple task categories, this study aims to contribute toward more reliable, effective, and responsible use of LLMs in data visualization applications.

Overall, this review highlights the strong potential of artificial intelligence in supporting and automating visualization-related tasks. However, continued investigation is necessary to improve the accuracy, consistency, and reliability of LLMs in practical, real-world visualization scenarios.

1.2 Motivation and Objectives

The primary goal of this project is to **evaluate the performance and accuracy of Large Language Models (LLMs) in interpreting standard charts**. While LLMs are increasingly used to support data analysis, their reliability in chart-related tasks such as chart-based question answering, visual encoding interpretation, and detection of misleading charts presents opportunities for further exploration. Understanding the limitations and common failure patterns of these models is essential for their effective and responsible application in visualization-driven workflows. By systematically analyzing model responses across multiple chart-based tasks, this project aims to provide insights into the strengths, weaknesses, and accuracy of contemporary LLMs, establishing baseline performance benchmarks and guiding future improvements.

The main objectives of this final-year project are as follows:

1. To **evaluate and compare the performance** of selected commercial and open-source LLMs, including **ChatGPT, Gemini, Claude, and Mistral**, across multiple chart-related task types.
2. To assess the ability of LLMs **to accurately interpret visual encoding elements** in standard charts.
3. To evaluate the performance of LLMs in **answering analytical questions** based on chart images.
4. To examine the capability of LLMs to **detect misleading or incorrect visualizations**.
5. To analyze **common error patterns** exhibited by LLMs when interpreting and reasoning about charts.
6. To **identify the strengths and limitations** of each evaluated LLM in handling graphical reasoning tasks.

1.3 Project Scope

The scope of this project focuses on evaluating the performance of multiple Large Language Models (LLMs) in chart-related tasks and identifying errors. The project includes:

- Evaluation of four LLMs: ChatGPT[1], Gemini[2], Claude[3], and Mistral[4].
- Tasks performed:
 - **Visual Encoding Interpretation:** answering questions based on visual encoding elements to check understanding of colors, position, shape and length.
 - **Chart-Based Question Answering:** answering questions based on chart images.
 - **Detection of Misleading Charts:** identifying charts with misleading or incorrect information.

- Organization of charts (e.g., bar, line, pie) into predefined categories (Common & Advanced) for systematic evaluation.
- Evaluation of LLM responses as correct, partially correct, or incorrect.
- Collection and classification of common error types for each model and task.

By focusing on reasoning, interpretation, and error analysis, this project provides a structured understanding of LLM performance in data-driven analytical tasks, highlighting both their current capabilities and potential for supporting real-world data analysis.

Chapter 2

Literature Review

Data visualization is a key approach for representing information graphically, supporting understanding and analysis. Common chart types such as bar charts, line charts, scatter plots, and pie charts are widely used for their clarity and effectiveness [5] [6]. Proper use of visual encodings, including position, color, size, and shape, is critical for accurate interpretation [7]. Libraries like Vega-Lite and D3.js have simplified visualization creation, but they require coding expertise, which can be a barrier for many users [8] [9]. Large Language Models (LLMs) such as ChatGPT, Gemini, and Claude have emerged as versatile tools capable of generating structured outputs and assisting with data tasks [10] [11]. Recent work demonstrates their potential for visualization applications: LLM4Vis showed that LLMs can produce visualization specifications from prompts [12], while AdaVis proposed adaptive chart recommendations based on user intent [13]. Despite these advances, LLMs still struggle with accuracy and consistency, especially for complex charts or multi-dimensional visualizations [14]. Beyond chart generation, interpreting visual data is a significant challenge. Studies like ChartQA have highlighted that models often make errors in extracting values, comparing data points, and reasoning about visual encodings [15]. Overall, while LLMs demonstrate strong potential for automating visualization tasks, gaps remain in understanding their limitations. This motivates a systematic evaluation of multiple LLMs across diverse chart-related tasks, aiming to assess the accuracy, reliability, and performance of these models in data reasoning and analysis.

Chapter 3

Methodology and Implementation

3.1 Overview of Methodology

This project adopts a systematic methodology to evaluate the performance of contemporary Large Language Models (LLMs) across a set of **predefined chart-related** tasks and systematically identifying common errors.

This study evaluates four widely used LLMs: ChatGPT, Gemini, Claude, and Mistral are evaluated. The methodology consists of designing standardized tasks, executing them consistently across models, and recording model outputs for analysis. The evaluation focuses on **identifying accuracy, inconsistencies, and recurring error patterns** to assess their ability **to interpret, reason about, and critique standard visualizations**.

Each task is designed using a category-based approach, where charts are grouped into specific categories based on chart type and visual characteristics. For each category, a set of labelled questions is created to assess different aspects of model understanding and reasoning.

3.2 Selected Large Language Models (LLMs)

These models are chosen due to their popularity, accessibility, and claimed capabilities in data reasoning and analysis.

The following LLMs are selected for evaluation:

Table 3.1: Evaluated Large Language Models and Configurations

LLMs	Version/Model	Developed By
ChatGPT	Model GPT-5.2	OpenAI
Claude	Model Claude Sonnet 4.5	Anthropic
Gemini	Model Gemini 2.5 Flash	Google
Mistral	Model Mistral-Large-2407	Mistral AI

Each model is tested using the same dataset and prompts to ensure a consistent baseline for comparison.

3.3 Data Collection

A diverse set of standard charts (bar, line, pie, scatter, etc.) is curated from public datasets and synthetically generated examples [16–34].

The charts are carefully selected to ensure coverage across a range of analytical tasks.

3.4 Tasks Definition

The performance of Large Language Models (LLMs) in chart-related tasks is systematically evaluated to identify common errors and limitations. To achieve this, the evaluation has been divided into multiple tasks, each targeting a specific aspect of data visualisation and analysis. The tasks are designed to cover chart interpretation and reasoning, ensuring a comprehensive assessment of the models’ abilities. This includes **understanding and applying visual encoding principles, interpreting data accurately from charts, and detecting potential misleading or false visualisations**. By defining these tasks clearly, the study aims to systematically measure model performance and document errors.

The following tasks are defined for evaluation of Large Language Models (LLMs):

3.4.1 Task 1: Visual Encoding Interpretation

This task evaluates whether Large Language Models (LLMs) correctly interpret visual elements such as colors, position, shape and length from the charts given to LLMs. Questions are designed to test the model’s ability to associate visual encodings with the correct categories or values. The objective is to assess the ability of Large Language Models (LLMs) to understand and reason about visual encoding elements in charts.

3.4.2 Task 2: Chart-Based Question Answering

Large Language Models (LLMs) are given chart images and asked questions related to the values, trends, and comparisons shown in the charts. Model responses are assessed for correctness and consistency. The objective is to evaluate the ability of LLMs to accurately interpret charts and answer questions about data values and trends.

3.4.3 Task 3: Detection of Misleading Charts

Large Language Models (LLMs) are presented with charts that contain intentional flaws, such as wrong color palette used or misleading scales. The task assesses the model’s ability to identify and explain potential issues that could lead to misinterpretation. The objective is to evaluate the ability of LLMs to detect misleading information or inconsistencies in charts.

3.5 Charts Categorisation

To systematically evaluate Large Language Models (LLMs) performance across different visualization types, the charts used in this study are categorized based on their structural complexity and common usage in practice. Prior visualization research has shown that certain chart types, such as bar charts, line charts, and pie charts, are more frequently used and easier to interpret [35].

Building on this insight, the charts in this study are organized into **two categories: Common Charts**, which are widely used and relatively straightforward to interpret,

and **Advanced Charts**, which feature more complex layouts or multi-dimensional encodings, presenting greater analytical challenges. This categorization enables a structured assessment of Large Language Models (LLMs) capabilities across varying levels of visual and data reasoning complexity.

Table 3.2: Chart Categories Used in Task 1

S.no	Common Chart	Advanced Chart
1	Bar Chart	Radar Chart
2	Line Chart	Stream Graph
3	Histogram	Tidy Tree
4	Scatter Plot	Treemap
5	Pie Chart	Candlestick
6	Donut Chart	Sankey Diagram
7	Box Plot	Parallel Coordinates
8	Heatmap	Sunburst Chart
9	Stacked Area Chart	Geospatial Chart
10	Grouped Bar Chart	Network Graph

Table 3.3: Chart Categories Used in Task 2

S.no	Common Chart	Advanced Chart
1	Bar Chart	Radar Chart
2	Line Chart	Stream Graph
3	Histogram	Tidy Tree
4	Scatter Plot	Treemap
5	Pie Chart	Candlestick
6	Donut Chart	Sankey Diagram
7	Box Plot	Parallel Coordinates
8	Heatmap	Sunburst Chart
9	Stacked Area Chart	Geospatial Chart
10	Grouped Bar Chart	Network Graph

Table 3.4: Chart Categories Used in Task 3

S.no	Common Chart	Advanced Chart
1	Bar Chart	Radar Chart
2	Line Chart	Stream Graph
3	Histogram	Treemap
4	Scatter Plot	Sankey Diagram
5	Pie Chart	Parallel Coordinates
6	Donut Chart	Sunburst Chart
7	Heatmap	Geospatial Chart
8	Stacked Area Chart	Network Graph
9	Grouped Bar Chart	
10	Stacked Bar Chart	

This classification allows for a comparative evaluation of Large Language Models (LLMs) performance on both standard and challenging visualizations. Tasks are applied to all charts to systematically capture model strengths and limitations across different levels of complexity.

3.6 Tasks Design

For each task, charts are grouped into predefined categories based on chart task and visual characteristics. Each category contains a set of labelled questions designed to evaluate specific aspects of model performance.

3.6.1 Task 1: Visual Encoding Interpretation

This task evaluates the ability of Large Language Models (LLMs) to interpret visual encoding elements in standard charts. Visual encoding refers to how data is represented graphically through attributes such as color, position, shape, and length. The objective is to assess whether Large Language Models (LLMs) can correctly identify and reason about these elements when answering questions based on chart images.

Four key categories of visual encoding are considered in this task:

Table 3.5: Visual Encoding Categories for Task 1

S.no	Category
1	Color Hue
2	Position
3	Shape
4	Length

1. Color (Hue)

This category evaluates whether Large Language Models (LLMs) can correctly identify and interpret the use of color in representing different data categories.

Objective: To determine whether Large Language Models (LLMs) can accurately recognize colors and associate them with the corresponding data categories.

Example Question: Which color represents the 16–20 age group?

2. Position

This category focuses on how data values are mapped to axes (e.g., x-axis and y-axis) in a chart.

Objective: To assess whether Large Language Models (LLMs) can correctly identify which axis represents specific variables or values (e.g., frequency, time, categories).

Example Question: Which axis shows the number of people?

3. Shape

This category examines whether Large Language Models (LLMs) can recognize

the shapes used to represent data points or groups within a chart.

Objective: To evaluate whether Large Language Models (LLMs) can correctly identify the graphical shapes used in the visualization (e.g., bars, points, markers).

Example Question: What shape is used to represent each age group?

4. Length (Size/Height)

This category evaluates how well Large Language Models (LLMs) interpret size-based encodings, such as the height or length of bars in a chart.

Objective: To determine whether Large Language Models (LLMs) can accurately compare values based on visual length or height.

Example Question: What is the length of the bar representing the age group with the second-highest value?

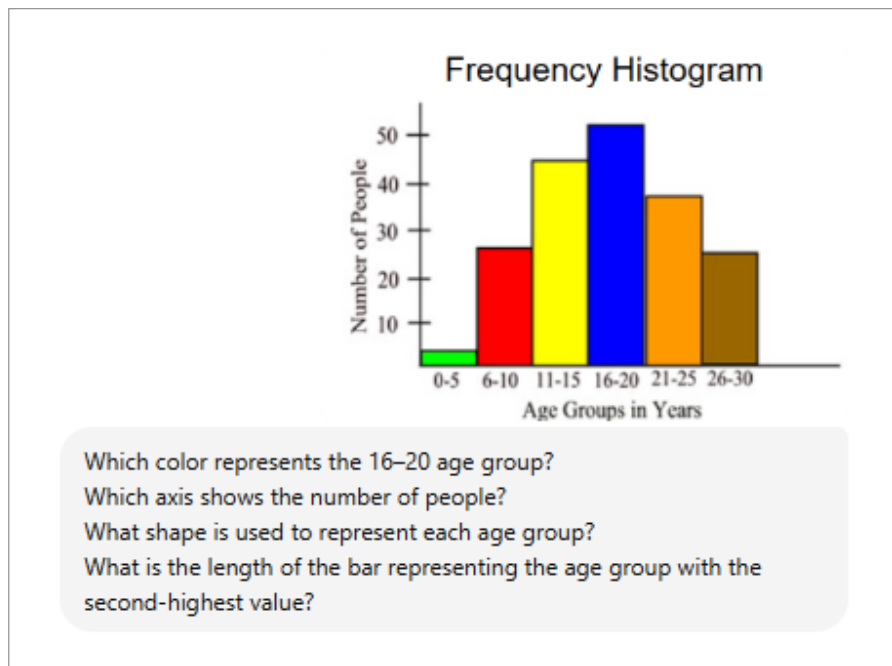


Figure 3.1: Example of visual encoding questions provided to ChatGPT for Task 1 .

3.6.2 Task 2: Chart-Based Question Answering

This task evaluates the ability of Large Language Models (LLMs) to perform analytical reasoning based on chart images. Unlike Task 1, which focuses on interpreting individual visual encoding elements, this task assesses whether LLMs can extract insights, compare values, and reason about the data presented in visual form.

The charts used in this task are categorized into common charts (e.g., bar charts, line charts, pie charts) and advanced charts (e.g., tidy tree, radar charts), allowing evaluation across varying levels of complexity. Large Language Models (LLMs) are provided with chart images and asked a series of structured questions designed to test different types of analytical reasoning.

For each chart, Large Language Models (LLMs) are asked questions across the following categories:

Table 3.6: Chart Based Question Answering Categories for Task 2

S.no	Category
1	Highest Value Identification
2	Lowest Value Identification
3	Direct Value Retrieval
4	Similarity Detection
5	Ranking
6	Difference Between Values
7	Counting-Based Reasoning

1. Highest Value Identification

This category evaluates whether Large Language Models (LLMs) can identify the maximum value represented in a chart.

Objective: To determine whether Large Language Models (LLMs) can accurately recognize the highest data point based on the visual representation.

Example Question: Which daily activity has the highest percentage in the chart?

2. Lowest Value Identification

This category evaluates whether Large Language Models (LLMs) can identify the minimum value in a chart.

Objective: To assess whether Large Language Models (LLMs) can correctly determine the lowest data point from the visual representation.

Example Question: Which activity has the lowest percentage?

3. Direct Value Retrieval

This category evaluates whether Large Language Models (LLMs) can extract specific values directly from a chart without requiring additional reasoning.

Objective: To determine whether models can accurately read and report exact data values from the visualization.

Example Question: What percentage of time is spent watching TV?

4. **Similarity Detection**

This category evaluates whether Large Language Models (LLMs) can identify data points with similar or closely related values.

The objective is to assess whether Large Language Models (LLMs) can compare values and recognize similarities within the chart.

Example Question: Which two categories have similar values?

5. **Ranking**

This category evaluates whether Large Language Models (LLMs) can correctly order data points based on their values.

Objective: To determine whether Large Language Models (LLMs) can rank categories in ascending or descending order using the visual information provided.

Example Question: What is the rank of “Sleep” among all activities?

6. **Difference Between Values**

This category evaluates whether Large Language Models (LLMs) can compute or estimate the difference between two data points.

Objective: To assess whether Large Language Models (LLMs) can perform comparative reasoning based on the values shown in the chart.

Example Question: What is the difference between the highest and lowest activity percentages?

7. **Counting-Based Reasoning**

This category evaluates whether Large Language Models (LLMs) can count the number of data points that satisfy a given condition.

Objective: To determine whether Large Language Models (LLMs) can accurately apply conditions and perform counting based on the chart data.

Example Question: How many activities have more than 20

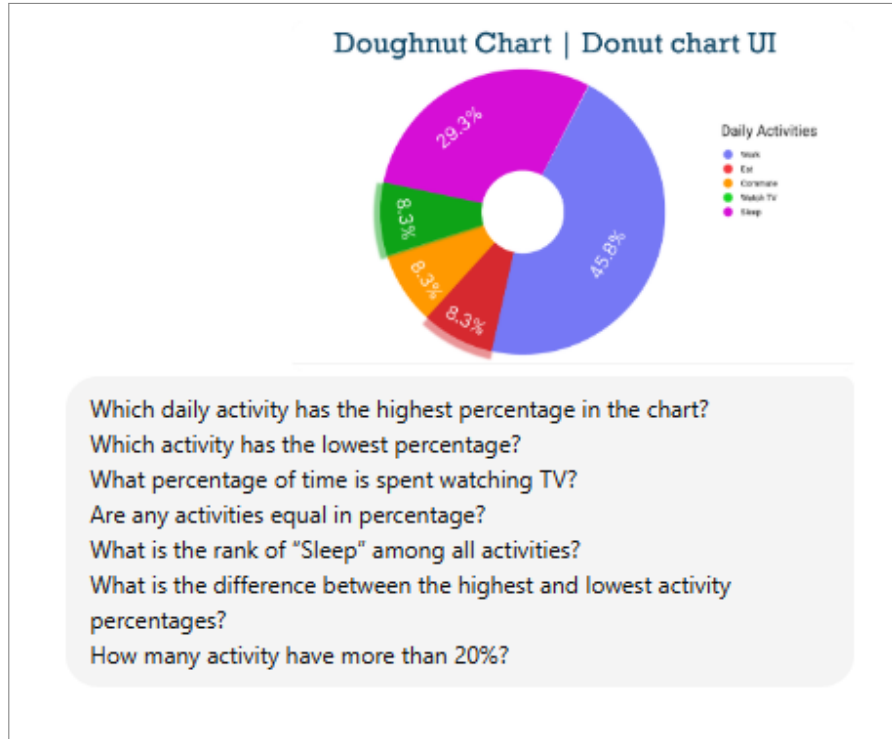


Figure 3.2: Example of chart based questions provided to ChatGPT for Task 2.

3.6.3 Task 3: Detection of Misleading Charts

This task evaluates the ability of Large Language Models (LLMs) to identify and explain misleading or incorrectly designed visualizations. Charts used in this task are intentionally created with common visualization flaws to assess whether LLMs can detect issues and provide accurate reasoning.

While Task 1 focuses on interpreting visual encoding elements and Task 2 focuses on analytical reasoning, Task 3 evaluates critical reasoning, requiring models to assess the correctness and effectiveness of a chart based on established data visualization principles.

Charts are designed with specific categories of misleading elements, as described below:

Table 3.7: Misleading Charts Categories for Task 3

S.no	Category
1	Choice of Chart
2	Color Mess
3	Overplotting / Overcrowding
4	Incorrect Size Encoding
5	3D Distortion
6	Missing / Ambiguous Legend

1. Choice of Chart

This category evaluates whether Large Language Models (LLMs) can identify the inappropriate use of chart types for representing given data.

Objective: To determine whether LLMs can recognize when a chart type does not effectively communicate the underlying data.

Example Question: Are there any flaws in the chart? Detect what is wrong and explain why.



Figure 3.3: Example of a pie chart illustrating the inappropriate choice of chart for this type of visualization provided to ChatGPT for Task 3.

2. Color Mess (Color Misuse)

This category evaluates whether Large Language Models (LLMs) can detect poor or misleading use of color, such as excessive colors, inconsistent color mapping,

or lack of distinction between categories.

Objective: To assess whether LLMs can identify how color choices impact clarity and interpretation.

Example Question: Are there any flaws in the chart? Detect what is wrong and explain why.

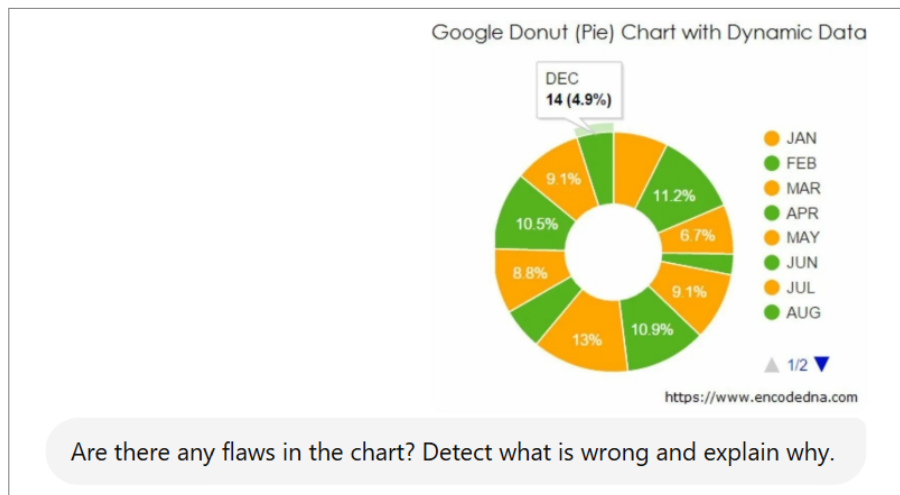


Figure 3.4: Example of a pie chart with color mess error provided to ChatGPT for Task 3.

3. Overplotting / Overcrowding

This category evaluates whether Large Language Models (LLMs) can identify charts where too much data is displayed, leading to clutter and reduced readability.

Objective: To determine whether LLMs can recognize when a visualization becomes difficult to interpret due to excessive information.

Example Question: Are there any flaws in the chart? Detect what is wrong and explain why.

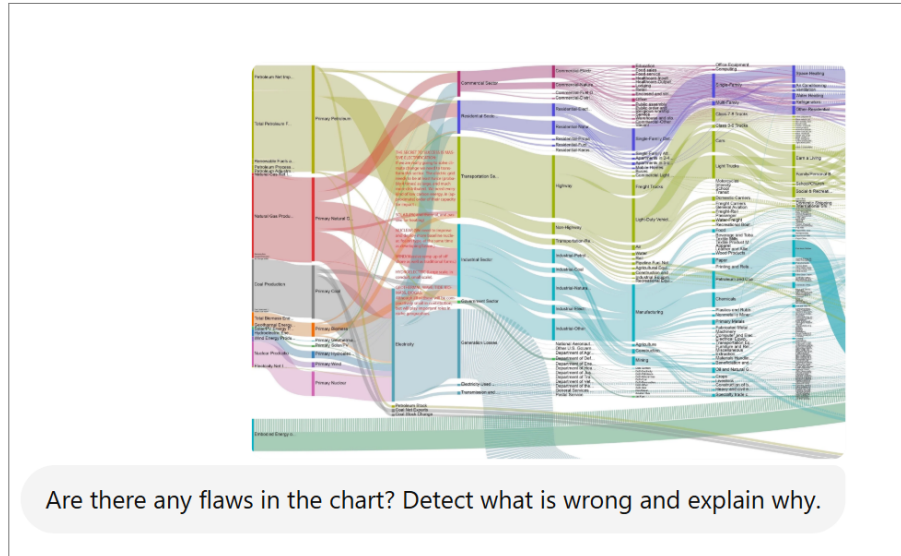


Figure 3.5: Example of a sankey diagram with excessive data density error for Task 3.

4. Incorrect Size Encoding

This category evaluates whether Large Language Models (LLMs) can detect inconsistencies between data values and their visual representation in terms of size, such as incorrect bar heights or area proportions.

Objective: To assess whether LLMs can identify mismatches between the data and its visual encoding.

Example Question: Are there any flaws in the chart? Detect what is wrong and explain why.



Figure 3.6: Example of an incorrect pie chart with size encoding error (exceeding 100%) provided to ChatGPT for Task 3.

5. 3D Distortion

This category evaluates whether Large Language Models (LLMs) can identify distortions caused by the use of 3D (Three Dimensional) effects in charts, which may misrepresent data proportions.

Objective: To determine whether Large Language Models (LLMs) can recognize how 3D visualizations can introduce bias or misinterpretation.

Example Question: Are there any flaws in the chart? Detect what is wrong and explain why.

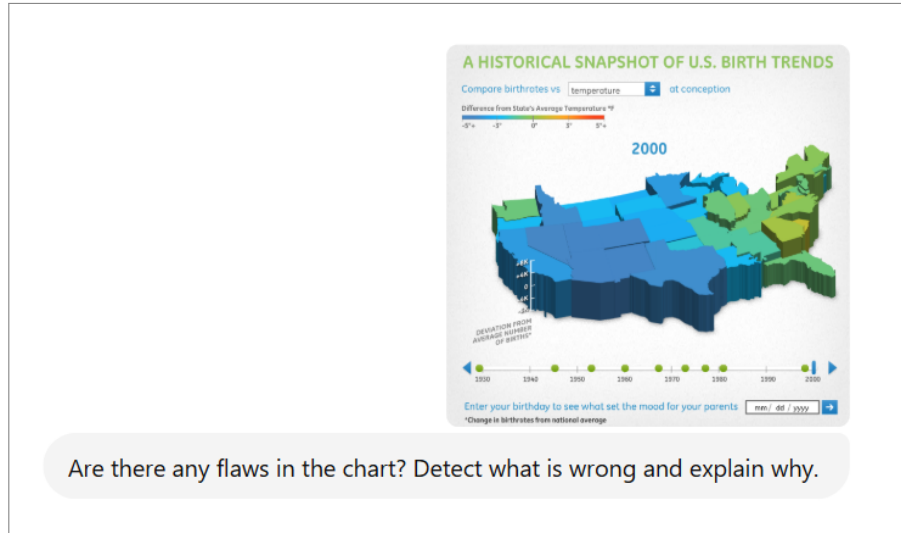


Figure 3.7: Example of a geospatial chart with 3D distortion provided to ChatGPT for Task 3.

6. Missing or Ambiguous Legend

This category evaluates whether Large Language Models (LLMs) can detect the absence of a legend, incomplete or unclear labeling, which makes the chart difficult to interpret.

Objective: To assess whether Large Language Models (LLMs) can identify missing contextual information required for understanding the visualization.

Example Question: Are there any flaws in the chart? Detect what is wrong and explain why.

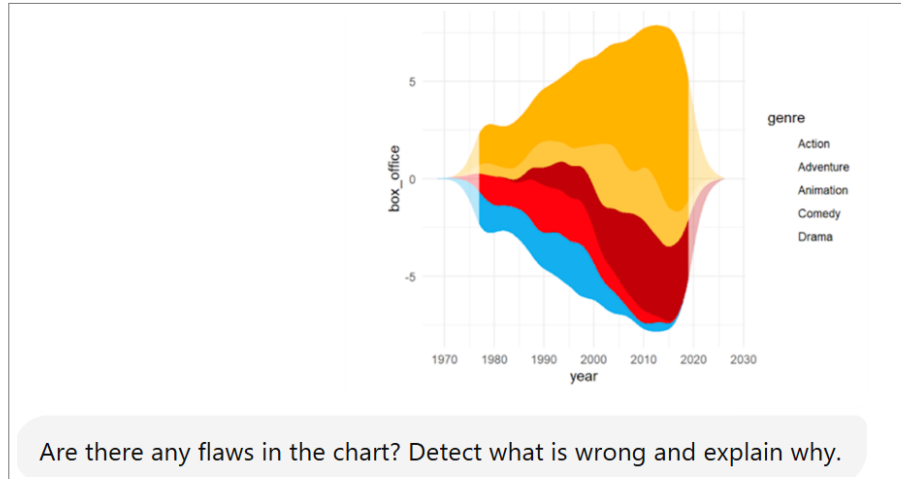


Figure 3.8: Example of a streamgraph with unclear legend provided to ChatGPT for Task 3.

3.7 Expected Output

For **Task 1: Visual Encoding Interpretation**, the expected output focuses on the accurate identification and interpretation of key visual encoding elements within charts. This includes correctly recognizing components such as color, position, shape, and length, as well as understanding chart elements like axes and labels. The responses should demonstrate a clear understanding of how data is represented visually and provide correct answers based on these encoding principles.

For **Task 2: Chart-Based Question Answering**, the expected output involves providing accurate answers to analytical questions derived from chart data. The responses should reflect a correct interpretation of the visual representation, including the ability to identify trends, compare values, and understand relationships within the data. Additionally, answers are expected to demonstrate logical and consistent reasoning based on the information presented in the chart.

For **Task 3: Detection of Misleading Charts**, the expected output requires the identification of misleading, incorrect, or poorly designed elements within visualizations. Responses should not only highlight these issues but also provide clear and well-reasoned explanations to justify why they are misleading. The answers are expected to

align with established data visualization principles and demonstrate critical evaluation of chart design and effectiveness.

Chapter 4

Experiments

This chapter presents the experimental design used to evaluate the baseline performance of multiple Large Language Models (LLMs) on chart-related tasks. For each task, a standardized set of chart images was prepared and categorized based on chart type and category. A consistent set of questions was applied across all evaluated models to ensure fairness and comparability.

All experiments were conducted using baseline prompts without any prompt optimization. Identical inputs, including chart images and question sets, were provided to each model. Model responses were systematically recorded for subsequent analysis.

The experimental design aims to enable a structured comparison of model performance while identifying recurring error patterns across tasks and chart categories.

4.1 Experimental Setup

To account for variability in LLMs outputs, each question in Tasks 1, 2, and 3 was presented five times using identical prompts. The response that appeared most frequently across these repetitions was selected as the majority-vote response and used for evaluation. This approach ensures that the evaluation reflects consistent model behavior rather than isolated or stochastic outputs.

All LLMs interactions were recorded in a structured spreadsheet [36], including

prompts, responses, repetition counts, and final majority-vote answers. This enabled systematic comparison across models, tasks, and chart categories.

4.2 Task Execution

For **Task 1 (Visual Encoding Interpretation)**, chart images sourced from publicly available datasets were used as input. Each chart type was paired with a set of four questions per category designed to assess the model’s understanding of visual encoding elements such as color, position, shape, and length. Each question was repeated five times, and the majority-vote response was used to evaluate the model’s ability to accurately interpret visual features.

Table 4.1: Number of chart questions used for Task 1 (Visual Encoding Interpretation).

Chart No.	Chart Category	No. of Chart Types	No. of Categories	Questions per Category	Total Questions
1	Common	10	4	5	200
2	Advanced	10	4	5	200
Total					400

For **Task 2 (Chart-Based Question Answering)**, chart images sourced from publicly available datasets were used as input. For each chart type, a set of seven questions categories was designed to evaluate different aspects of analytical reasoning, including highest and lowest value identification, direct value retrieval, similarity detection, ranking, difference calculation, and counting. Each question was repeated five times per model using identical prompts, and the majority-vote response was recorded. Responses were evaluated based on their correctness and consistency with the information presented in the chart.

Table 4.2: Number of chart questions used for Task 2 (Chart-Based Question Answering).

Chart No.	Chart Category	No. of Chart Types	Categories	Questions per Category	Total Questions
1	Common	10	7	5	350
2	Advanced	10	7	5	350
Total			14		700

For **Task 3 (Detection of Misleading Charts)**, a targeted set of chart types and pre-defined error categories was selected. For each category, four intentionally flawed charts were sourced from online datasets or created, incorporating common visualization issues such as inappropriate choice of chart type, poor use of color, overplotting, incorrect size encoding, 3D distortion, and missing or ambiguous legends. These chart images were provided as input to the Large Language Models (LLMs), which were prompted to identify and explain any misleading elements. Each analysis prompt was repeated five times, and the majority-vote response was used for evaluation to reduce inconsistencies in model outputs.

Table 4.3: Number of chart questions used for Task 3 (Detection of Misleading Charts).

Chart No.	Chart Category	No. of Chart Types	Categories per Chart	Questions per Category	Total Questions
1	Common	10	6	4	240
2	Advanced	8	6	4 ¹	198
Total		18			438

Across all tasks, responses were systematically recorded to enable consistent error tracking, comparison across models, and identification of recurring failure patterns. This structured execution provides a robust experimental foundation for analyzing the strengths and limitations of Large Language Models (LLMs) in chart-related tasks.

¹One chart type contains 5 questions instead of 4.

Chapter 5

Evaluation

5.1 Evaluation Metrics

This chapter focuses on assessing baseline performance and identifying error patterns. Hence, each model response was classified into one of the following categories:

- Correct
- Partially correct
- Incorrect

For each task and chart category, responses were classified as correct, partially correct, or incorrect. To systematically evaluate the performance of the large language models (LLMs) across different tasks and chart types, a combination of quantitative and qualitative metrics was defined.

Three key evaluation metrics are employed: **accuracy, partial correctness, and consistency**. Together, these metrics provide a structured framework to assess both the accuracy and reliability of the model outputs, ensuring the evaluation is comprehensive, consistent, and reproducible.

5.1.1 Accuracy

Accuracy measures the proportion of responses that are fully correct for each task, match the expected output and reflects the model’s ability to produce reliable and precise outputs.

For **Task 1 (Visual Encoding Interpretation)**, accuracy is determined by the correct identification and interpretation of visual encoding elements such as color, position, size, and shape. A response is considered accurate when the model correctly associates these visual features with the corresponding data values or categories.

As shown in Figure 5.1 and Table 5.1, all LLMs correctly identified the color encoding of the specified category. The responses are considered accurate, as the identified colors either exactly match or closely correspond to the actual chart colors (e.g., the chart uses blue, and the LLMs also identified it as blue), demonstrating a correct interpretation of visual encoding elements.

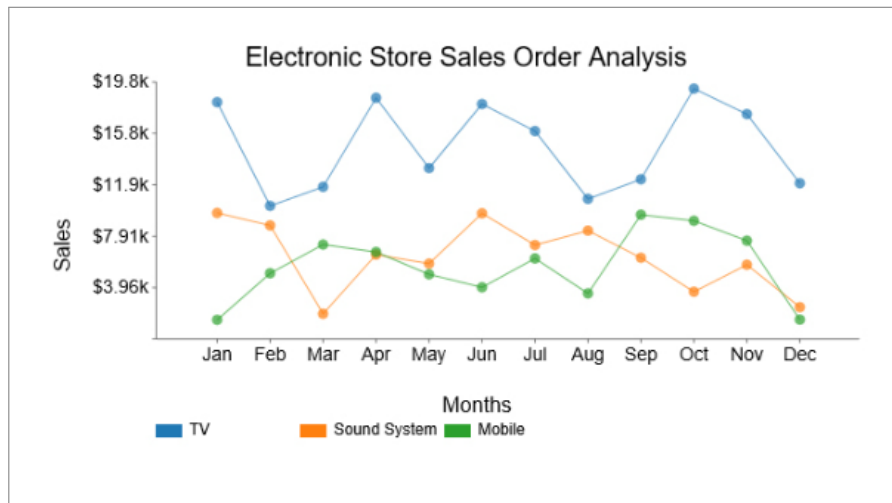


Figure 5.1: Example of Line Chart used for Task 1 (Visual Encoding Interpretation).

Table 5.1: Example of a color visual encoding question for Task 1 (Visual Encoding Interpretation) alongside responses generated by LLMs.

Which color represents the sales trend for TVs?			
Expected Output: Blue			
ChatGPT	Gemini	Claude	Mistral
The sales trend for TVs is represented by the blue line.	Blue represents the sales trend for TVs.	The blue line represents the sales trend for TVs.	The blue line in the chart represents the sales trend for TVs.

For **Task 2 (Chart-Based Question Answering)**, accuracy refers to the ability of the model to correctly answer analytical questions based on the chart. This includes identifying values, recognizing trends, and performing comparisons in alignment with the data presented.

As shown in Figure 5.2 and Table 5.2, ChatGPT, Gemini, and Claude correctly identified the team with the highest score in Round 1, whereas Mistral failed to provide the correct answer. The responses from ChatGPT, Gemini, and Claude are considered accurate, as they align with the expected output. In contrast, Mistral’s response is classified as incorrect due to its failure to identify the correct data point.

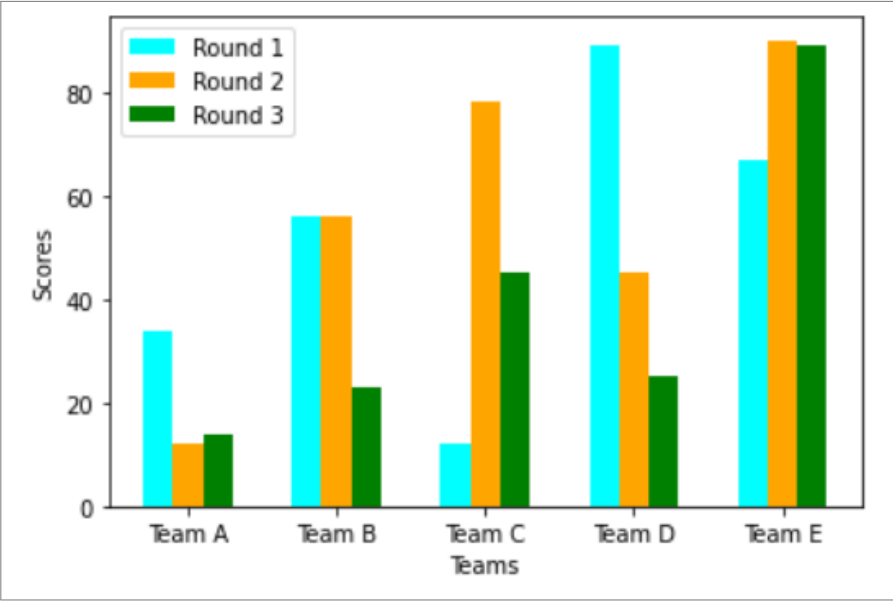


Figure 5.2: Example of Grouped Bar chart used for Task 2 (Chart-Based Question Answering)

Table 5.2: Example of Highest Value Identification question for Task 2 (Chart-Based Question Answering) alongside responses generated by LLMs.

Which team has the highest score in Round 1?			
Expected Output: Team D			
ChatGPT	Gemini	Claude	Mistral
From the chart, Team D has the highest score in Round 1.	Team D has the highest score in Round 1.	Team D has the highest score in Round 1, with the cyan bar reaching approximately 88 points.	The bar chart shows that Team E has the highest score in Round 1, with a score of around 80.

For **Task 3 (Detection of Misleading Charts)**, accuracy is defined by the model’s ability to correctly identify misleading or incorrect elements within a chart and provide valid explanations. Responses are considered accurate when both the issue and its

impact on interpretation are correctly described.

Similarly, as illustrated in Figures 5.3 and 5.3, all LLMs correctly identified that the scatter plot is missing a legend. These responses are considered accurate, as the models successfully detected the issue and provided appropriate reasoning explaining why the absence of a legend may lead to misinterpretation of the chart.

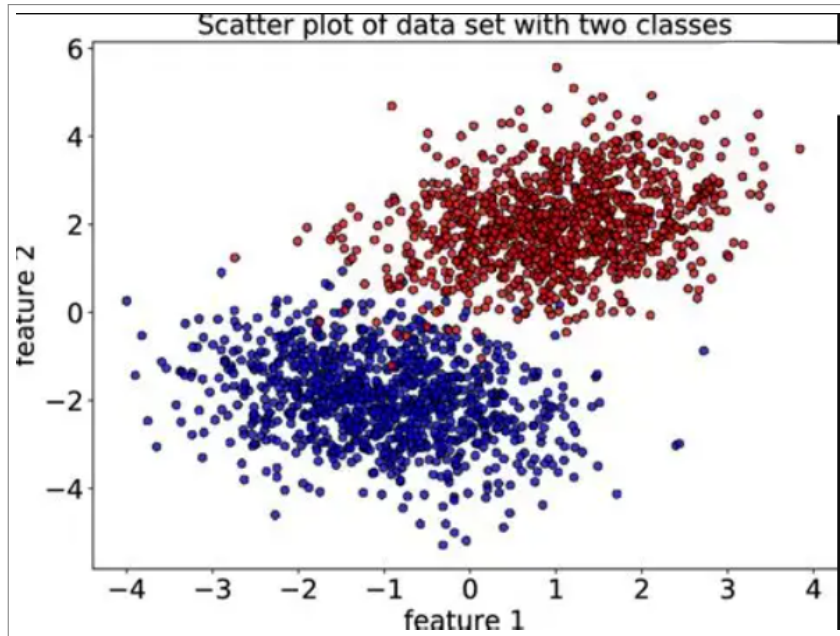


Figure 5.3: Example of scatter plot with missing legend used for Task 3 (Detection of Misleading Charts).

Are there any flaws in the chart? Detect what is wrong and explain why.
Expected Output: Missing Legend
ChatGPT • Missing legend: Red and blue classes are not labeled. • Poor class labeling: Title says “two classes” but classes are unnamed. • Overplotting: Many overlapping points make density unclear. • Limited context: Axes labeled generically as “feature 1” and “feature 2”. Gemini • Missing legend: No explanation of colors. • Overplotting: Dense clusters obscure individual points. • Lack of units: Axes generic with no physical meaning. • Class overlap: Red and blue points overlap significantly. • Poor margin usage: Points close to borders. Claude • Missing legend: Classes not labeled. • Overlapping points: Dense regions hard to interpret. • Generic title: ”Scatter plot of data set with two classes”. • No class labels anywhere. • Generic axis labels: Feature names are placeholders. • No gridlines: Hard to estimate point positions. Mistral • Overlapping points obscure distribution. • Lack of transparency: Points fully opaque. • No clear separation between classes. • Generic axes and title. • Missing legend: Red/blue colors not explained. • Suggested improvements: Use transparency, add legend, provide context, consider alternative visualizations, add gridlines.

Table 5.3: Responses of models for Task 3 (Detection of Misleading Charts), responses have been compacted for brevity.

5.1.2 Partial Correctness

Partial correctness captures responses that are generally correct but contain minor errors or omissions. This metric allows for a more nuanced evaluation of LLM performance beyond a strict binary classification.

The inclusion of partial correctness allows for a more nuanced evaluation, capturing cases where models demonstrate approximate reasoning without full precision.

In **Task 1 (Visual Encoding Interpretation)**, a response is considered partially correct when the model's output is close to the expected answer. For example, if the model identifies a color that is similar but not identical to the actual color, it is classified as partially correct.

As illustrated in Figures 5.4 and Table 5.4, the lowest grade interval in the histogram is represented using a deep reddish-orange color. However, Mistral described this color as red. Although the response is not fully accurate, it is classified as partially correct because the model demonstrates an approximate understanding of the visual encoding. Red and dark orange belong to closely related color hues and can be visually similar, particularly in the absence of precise color calibration or when subtle shade differences are difficult to distinguish. Therefore, the response reflects a reasonable interpretation of the color, despite lacking exact precision. Hence, this is marked as partially correct.

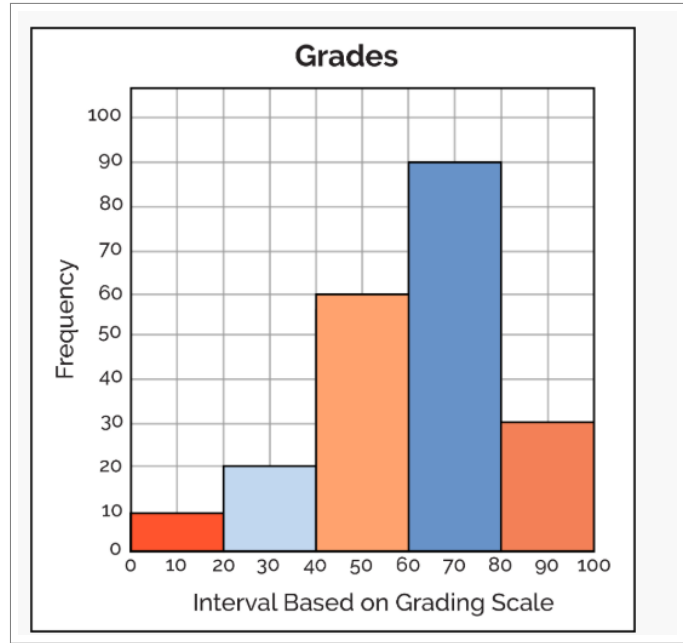


Figure 5.4: Example of histogram used for Task 1 (Visual Encoding Interpretation) provided to ChatGPT.

Question: Which color represents the bar for the lowest grade interval (0–20)?			
Expected Output: Dark Orange			
ChatGPT	Gemini	Claude	Mistral
The bar for the lowest grade interval 0–20 is the reddish-orange (bright red/orange) bar on the far left.	Based on the histogram provided, the bar for the lowest grade interval (0–20) is represented by the color orange (specifically a bright, reddish-orange).	The red/orange-red color represents the bar for the lowest grade interval (0–10), which has a frequency of 10.	In the histogram, the red bar represents the lowest grade interval (0–20).

Table 5.4: Responses of LLMs for Task 1 (Visual Encoding Interpretation).

For **Task 2 (Chart-Based Question Answering)**, responses are classified as partially correct when the model demonstrates an understanding of the chart but makes small numerical errors, incomplete comparisons, or minor misinterpretations of values while still reflecting the overall trend or intent of the question.

As shown in Figure 5.5 and Table 5.5, the LLMs were asked to identify the value of a point on the line chart corresponding to the 50–69 age group in 2021. The expected value is 165.

Gemini’s response is correct. ChatGPT provided a response indicating that the value is “above 150 and below 170,” which, although not exact, falls within the correct range. Similarly, Claude provided values close to the expected number. These responses are therefore considered partially correct, as they capture the correct range and relative magnitude despite lacking precise accuracy.

In contrast, Mistral reported a value of 180, which deviates significantly from the expected value and is therefore considered incorrect.

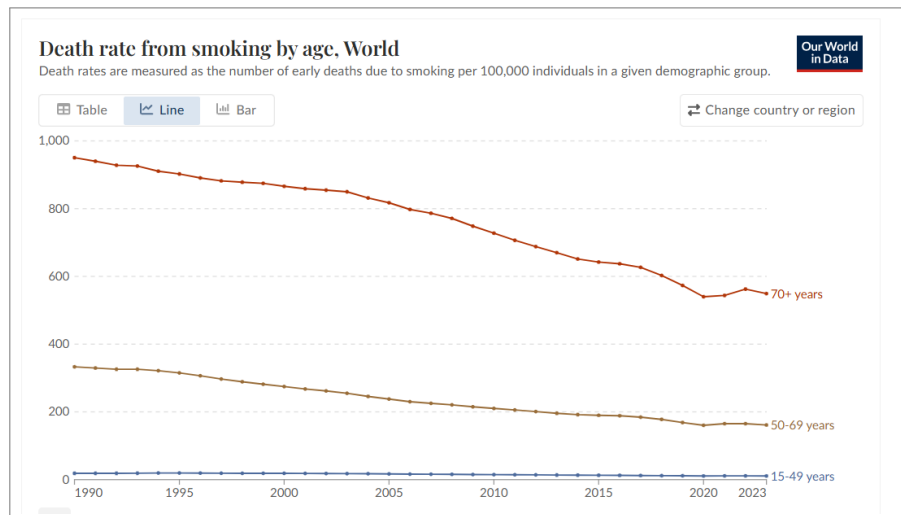


Figure 5.5: Example of Line chart used for Task 2 (Chart-Based Question Answering).

Table 5.5: Responses of LLMs for Task 2 (Chart-Based Question Answering).

What is the approximate value for the 50–69 years group in 2021?			
Expected Output: 165			
ChatGPT	Gemini	Claude	Mistral
From the chart, the 50–69 years line in 2021 sits just above the 150 mark and below 170.	Based on the chart, the approximate value for the 50–69 years group in 2021 is 165.	The approximate value for the 50–69 years group in 2021 is around 175 per 100,000 individuals.	The approximate death rate from smoking for the 50–69 years age group in 2021 is around 180 per 100,000 individuals.

For **Task 3 (Detection of Misleading Charts)**, responses are considered partially correct if the model successfully identifies at least one misleading aspect of the chart but fails to detect all introduced flaws or provides an incomplete or weak explanation of the identified issue.

As illustrated in Figure 5.6 and Table 5.6, the chart contains a critical flaw in the form of an incomplete legend, with some categories missing. Gemini and Claude correctly identified this flaw, and their responses are therefore considered correct. In contrast, Mistral and ChatGPT failed to detect the missing legend, so their responses are considered incorrect.

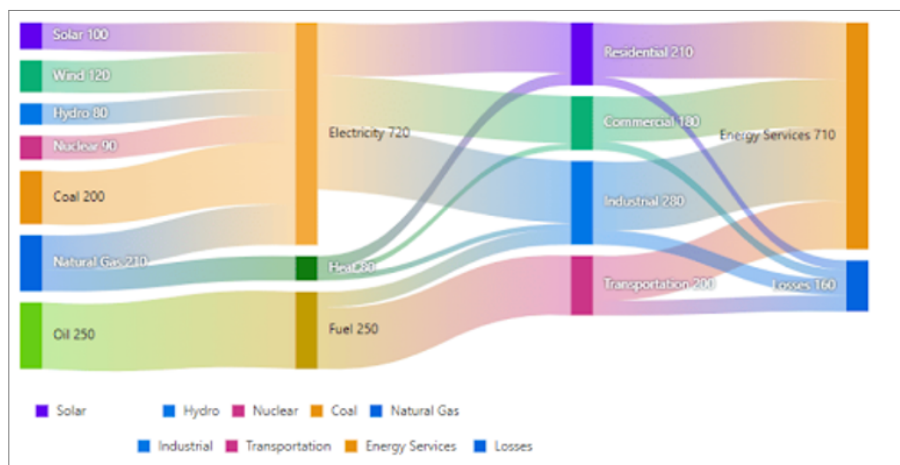


Figure 5.6: Example of Sankey Chart used for Task 3 (Detection of Misleading Charts) highlighting the incomplete legend issue.

Table 5.6: Example of a Sankey diagram Task 3 (Detection of Misleading Charts) alongside responses generated by LLMs.

Are there any flaws in the chart? Detect what is wrong and explain why.			
Expected Output: Incomplete Legend			
ChatGPT	Gemini	Claude	Mistral
• Energy doesn't balance: Inputs and outputs (e.g., "Electricity 720") don't clearly add up, breaking Sankey rules. • Double counting: Fuels like coal and gas appear in both electricity generation and direct use without clear separation. • Mixed categories: "Electricity" and "Fuel" are treated at the same level even though they're different types of energy. • Losses unclear: Only one "Losses" block is shown, hiding where energy is actually lost. • Confusing visuals: Colors and categories are inconsistent, making flows hard to follow.	The Sankey diagram contains several technical flaws: • Flow Imbalance: The sum of source inputs (1050) does not match the sum of final outputs (870), violating the principle of conservation. • Logical Errors: Solar and Wind are shown flowing into "Fuel," and Nuclear flows into "Heat," which are physically incorrect energy transitions. • Legend Mismatch: Several categories (Wind, Oil, Residential) are missing from the legend, and colors for "Natural Gas" and "Hydro" are nearly indistinguishable. • Missing Conversion Losses: Energy "Losses" (160) are only shown at the end-use stage, ignoring the significant generation losses typical in electricity production.	• Incomplete Legend — Wind, Oil, Residential, and Commercial nodes are missing from the legend. • Similar Colors — Too many shades of blue/green make flows hard to distinguish. • Overlapping Labels — Right-side labels like "Energy Services" and "Losses" are cramped. • Truncated Label — "Heat80" is missing a space, indicating a formatting error. • No Units — Numbers lack units (e.g., TWh, GW), making the data uninterpretable. • Value Imbalance — Input total (1,050) doesn't fully match output total (1,030), leaving a 20-unit gap unexplained.	1. Missing Flow Labels: Flows lack numerical values, making it hard to track exact energy distribution. 2. Color Confusion: Similar colors for different sources/sectors can cause misinterpretation. 3. No Units: Values lack units (e.g., MW, GW), reducing clarity. 4. Lack of Context: No time period or location is specified. 5. Inconsistent Flow Widths: Flow widths may not accurately reflect the data. 6. Overlapping Flows: Some flows overlap, making paths unclear.

5.1.3 Consistency

Consistency evaluates the repeatability of LLM outputs. For Tasks 2 through 4, each question was presented five times to account for potential variability in model responses. The majority-vote response was recorded as the representative output, and consistency was calculated as the proportion of repeated responses that matched this majority-vote output. This metric ensures that observed performance is reliable and not the result of random variation in model outputs.

As illustrated in Figure 5.7 and Table 5.7, this approach ensures that the observed performance reflects the models' true capabilities rather than random variation or session-specific differences.

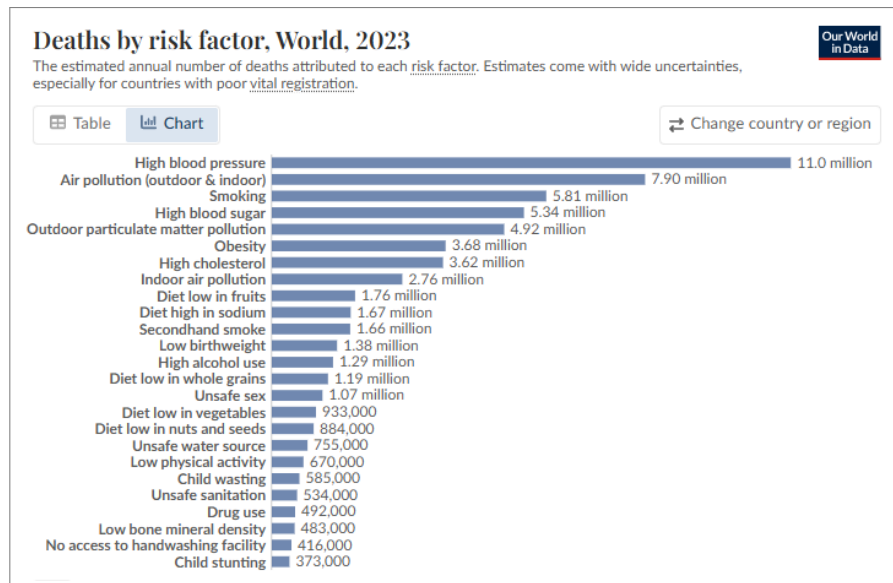


Figure 5.7: Example of a chart provided to LLMs for confirmation of responses. The chart was used to assess the reliability and repeatability of model outputs across multiple trials.

Table 5.7: Consistency Analysis of Model Responses Across Repeated Trials

Question: How many risk factors result in deaths more than 5 million?				
Expected Output: 4				
Gemini (Trial 1)	Gemini (Trial 2)	Gemini (Trial 3)	Gemini (Trial 4)	Gemini (Trial 5)
There are 4 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 4 risk factors that result in more than 5 million deaths.

The same chart and its associated question were presented to Gemini across five separate sessions. To ensure a consistent and representative evaluation, the response that appeared most frequently among these five repetitions was recorded as the majority-vote response. From Table 5.7, Gemini reported the answer “5” three times and “4” two times. Consequently, “5” is considered the majority-vote response for Gemini.

5.2 Results

5.2.1 Task 1: Visual Encoding Interpretation

Task 1 evaluates the ability of LLMs to interpret fundamental visual encoding elements, including color, position, shape, and length. A total of 400 questions were used, equally distributed across common and advanced chart types.

Chart No.	Chart Category	No. of Chart Types	No. of Categories	Questions per Category	Total Questions
1	Common	10	4	5	200
2	Advanced	10	4	5	200
Total					400

Table 5.8: Number of chart instances used for Task 1 (Visual Encoding Interpretation). Each chart type includes multiple instances selected to evaluate LLMs’ ability to answer visual encoding questions.

Overall, the results indicate strong performance across all evaluated models. ChatGPT achieved the highest number of correct responses, with 328, closely followed by Gemini (324) and Claude (305). In comparison, Mistral recorded a lower performance, with 293 correct responses.

LLMs	Correct Responses	Partially Correct Responses	Incorrect Responses
ChatGPT	328	18	54
Gemini	324	19	57
Claude	305	26	69
Mistral	293	18	89

Table 5.9: Task 1 evaluation results for LLMs. Responses are categorized as correct, partially correct, or incorrect based on answers by the LLMs.

A more detailed breakdown by encoding category reveals distinct performance patterns. All models performed exceptionally well in shape encoding, with near-perfect accuracy (e.g., ChatGPT and Claude both achieved 99 correct responses). This suggests that discrete and visually explicit features are easier for LLMs to interpret.

Performance in color encoding was also relatively strong, although minor inaccuracies were observed due to approximate color identification.

In contrast, position and length (size/height) encodings showed comparatively lower accuracy. Errors in these categories were more frequent, particularly in advanced charts where spatial relationships and relative magnitudes are less visually explicit. Length-based interpretation, which requires comparing bar heights or relative sizes,

was especially challenging, with Mistral recording the highest number of incorrect responses in this category.

Category	Evaluation	ChatGPT	Gemini	Claude	Mistral
Color Hue	Correct	83	84	78	73
	Partially Correct	6	5	12	6
	Incorrect	11	11	10	21
Position	Correct	77	77	73	68
	Partially Correct	6	4	6	6
	Incorrect	17	19	21	26
Shape	Correct	99	95	99	95
	Partially Correct	0	3	0	1
	Incorrect	1	2	1	4
Length (Height/Size)	Correct	69	68	55	57
	Partially Correct	6	7	8	5
	Incorrect	25	25	37	38

Table 5.10: Task 1 Category-wise evaluation of LLM responses on chart categories.

In Task 1, ChatGPT has the highest number of correct responses while Mistral has the least number of correct responses. Overall, the results suggest that LLMs are effective in interpreting simple and visually distinct encoding elements, but their performance declines when tasks require precise spatial reasoning or comparative analysis.

5.2.2 Task 2: Chart-Based Question Answering

Task 2 evaluates the ability of LLMs to perform analytical reasoning based on chart data. A total of 700 questions were used across seven question categories.

Chart No.	Chart Category	No. of Chart Types	Categories	Questions per Category	Total Questions
1	Common	10	7	5	350
2	Advanced	10	7	5	350
Total			14		700

Table 5.11: Number of chart instances used for Task 2 (Chart-Based Question Answering). Each chart type includes multiple instances selected to evaluate LLMs.

Overall, the results indicate strong performance across all evaluated models. Gemini (518) achieved the highest number of correct responses, followed by ChatGPT(467). In comparison, Claude and Mistral recorded a lower performance, with 398 and 359 correct responses respectively.

LLMs	Correct Responses	Partially Correct Responses	Incorrect Responses
ChatGPT	467	40	193
Gemini	518	45	137
Claude	398	63	239
Mistral	359	42	299

Table 5.12: Task 2 evaluation results for LLMs. Responses are categorized as correct, partially correct, or incorrect based on answers by the LLMs.

Across question categories, models performed best in highest value identification and similarity detection, where answers rely on identifying prominent visual patterns. Gemini consistently outperformed other models in these categories, indicating stronger pattern recognition capabilities.

However, performance declined significantly for more complex tasks such as ranking, difference calculation, and direct value retrieval. These tasks require precise numerical reasoning and careful interpretation of chart data. For example, Mistral showed particularly high error rates in ranking and difference-based questions, suggesting limitations in multi-step reasoning.

Partial correctness was more prominent in this task compared to Task 1, reflecting that models often captured general trends but failed to produce precise answers.

Category	Evaluation	ChatGPT	Gemini	Claude	Mistral
Highest Value Identification	Correct	78	85	74	64
	Partially Correct	5	4	4	4
	Incorrect	17	11	22	32
Lowest Value Identification	Correct	63	71	59	51
	Partially Correct	6	8	11	7
	Incorrect	31	21	30	42
Direct Value Retrieval	Correct	62	68	47	42
	Partially Correct	6	9	15	7
	Incorrect	32	23	38	51
Similarity Detection	Correct	71	72	61	60
	Partially Correct	5	4	5	3
	Incorrect	24	24	34	37
Ranking	Correct	63	78	49	47
	Partially Correct	6	4	7	5
	Incorrect	31	18	44	48
Difference between Values	Correct	62	67	49	43
	Partially Correct	7	9	13	13
	Incorrect	31	24	38	44
Counting-Based Reasoning	Correct	68	77	59	52
	Partially Correct	5	7	8	3
	Incorrect	27	16	33	45

Table 5.13: Task 2 Category-wise evaluation of LLM responses on chart categories.

Results indicate that models were highly accurate in extracting simple data trends and categorical comparisons, such as identifying the highest values. Partial correctness was observed when models failed to answer questions requiring extracting text from the chart and calculating differences.

Overall, the results demonstrate that while LLMs can effectively handle basic analytical queries, their ability to perform accurate numerical reasoning and multi-step analysis remains limited, especially for advanced charts.

5.2.3 Task 3: Detection of Misleading Visualisations

Task 3 evaluates the ability of LLMs to identify misleading or flawed visualizations. A total of 438 questions were used across multiple error categories.

Chart No.	Chart Category	No. of Chart Types	Categories per Chart	Questions per Category	Total Questions
1	Common	10	6	4	240
2	Advanced	8	6	4 ¹	198
Total		18			438

Table 5.14: Number of chart instances used for Task 3 (Detection of Misleading Charts). Each chart type includes multiple instances selected to evaluate LLMs’ ability to detect misleading or flawed visualizations.

Overall, the results indicate strong performance across all evaluated models. Gemini (350) achieved the highest number of correct responses, followed by Claude (346). In comparison, ChatGPT and Mistral recorded a lower performance, with 339 and 268 correct responses respectively.

LLMs	Correct Responses	Partially Correct Responses	Incorrect Responses
ChatGPT	339	19	80
Gemini	350	15	73
Claude	346	14	78
Mistral	268	10	160

Table 5.15: Task 3 evaluation results for LLMs. Responses are categorized as correct, partially correct, or incorrect based on identification of misleading chart elements and quality of explanations

The results vary notably across error categories. Models performed well in identifying overplotting, color issues, and 3D distortion issues, as these flaws are visually prominent and easier to detect.

However, performance declined in categories such as choice of chart and incorrect

¹One chart type contains 5 questions instead of 4.

size encoding, where errors require deeper conceptual understanding of visualization principles. For example, Mistral showed particularly high incorrect responses in incorrect size encoding and inappropriate chart selection errors, highlighting limitations in understanding appropriate visual representations.

Category	Evaluation	ChatGPT	Gemini	Claude	Mistral
Choice of Chart	Correct	43	30	40	27
	Partially Correct	5	4	4	2
	Incorrect	25	39	29	44
Color Mess	Correct	63	61	60	52
	Partially Correct	2	1	2	3
	Incorrect	8	11	11	18
Overplotting	Correct	66	65	63	69
	Partially Correct	4	5	1	1
	Incorrect	3	3	9	3
Incorrect Size Encoding	Correct	48	69	57	28
	Partially Correct	0	1	4	1
	Incorrect	25	3	12	44
3D Distortion	Correct	60	65	64	51
	Partially Correct	4	1	1	0
	Incorrect	9	7	8	22
Missing or Ambiguous Legend	Correct	59	60	62	41
	Partially Correct	4	3	2	3
	Incorrect	10	10	9	29

Table 5.16: Task 3 Category-wise evaluation of LLM responses on chart categories.

Overall, the performance of the LLMs varied considerably across different types of visualizations. Models were generally successful at detecting obvious flaws, such as overplotting and color misuse, but struggle with subtle or conceptually complex issues that require domain knowledge and critical reasoning.

The accuracy was highest for common chart types, such as basic bar charts, line charts, where the structure is familiar and the errors are visually obvious. In contrast, more advanced charts, particularly those with multiple layers of information, nested

hierarchies, or subtle data manipulations for e.g. streamgraph and treemaps, presented greater challenges.

5.3 Error Analysis

Across all three tasks, a consistent trend emerges in which model performance is strongly influenced by the complexity of the task and the level of reasoning required.

5.3.1 Task 1: Visual Encoding Interpretation

In Task 1, errors are primarily associated with limitations in precise visual perception. Models frequently approximate colors rather than identifying exact shades and encounter difficulty interpreting spatial relationships such as position and relative length.

Overall, the results indicate strong performance across all models, with accuracy exceeding 73% for all LLMs. ChatGPT achieved the highest accuracy (82.00%), closely followed by Gemini (81.00%), while Claude (76.25%) and Mistral (73.25%) exhibited comparatively lower performance.

Table 5.17: Performance of LLMs on Task 1 (percentage of correct, partially correct, and incorrect responses).

LLMs	% Correct	% Partially Correct	% Incorrect
ChatGPT	82.00%	4.50%	13.50%
Gemini	81.00%	4.75%	14.25%
Claude	76.25%	6.50%	17.25%
Mistral	73.25%	4.50%	22.25%

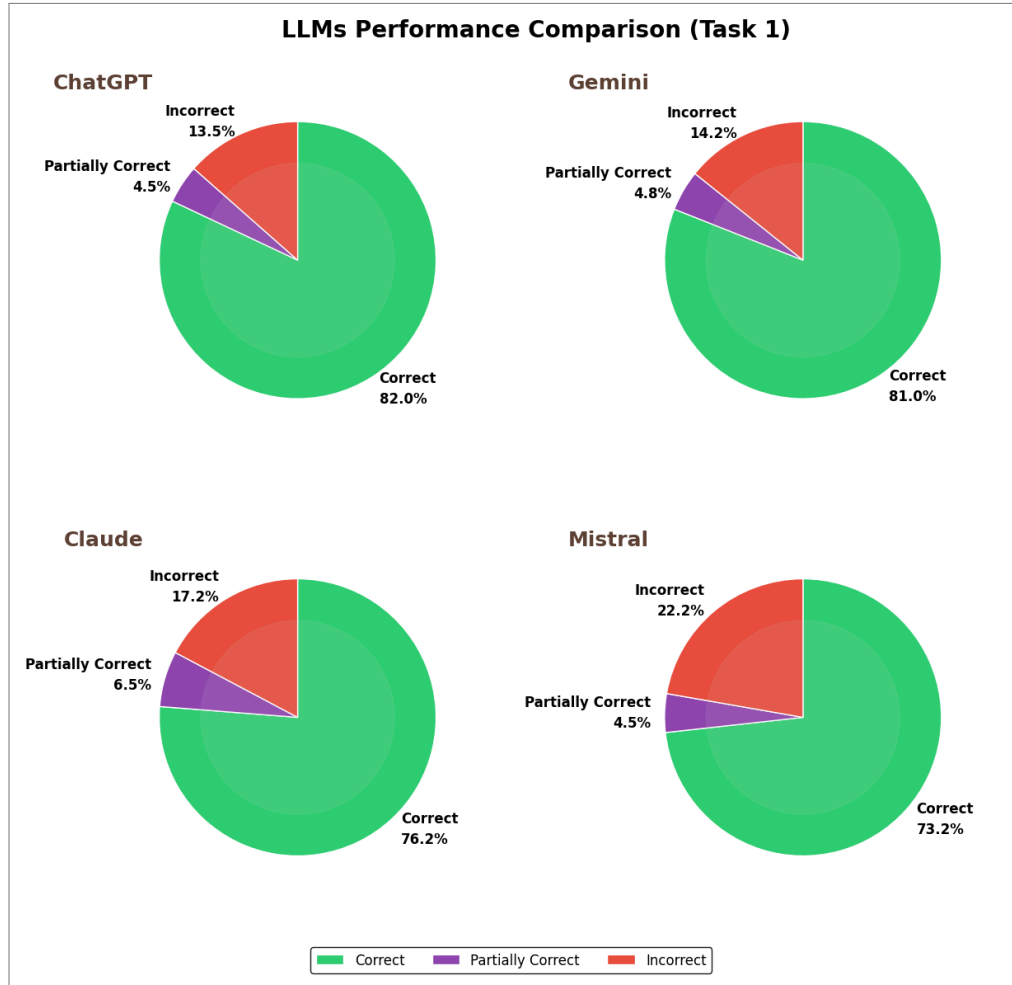


Figure 5.8: Visualisation of Task 1 performance of LLMs (percentage of correct, partially correct, and incorrect responses)

5.3.2 Task 2: Chart-Based Question Answering

In Task 2, errors are more pronounced and largely stem from limitations in numerical reasoning. Models often struggle with tasks requiring multi-step calculations, such as ranking or computing differences between values. Misinterpretation of axes and incorrect extraction of numerical values are also common sources of error.

For Task 2, LLMs generally demonstrated strong performance when answering chart-based questions for Common charts. Simple chart types such as bar charts, line charts, and pie charts were interpreted accurately, particularly for straightforward questions involving highest value retrieval. However, performance degraded significantly for

Advanced charts, including network graphs, tidy trees, and geospatial visualizations. Questions that required higher-level reasoning such as ranking data points or calculating differences between values posed greater challenges, even for Common charts. This suggests that while LLMs handle basic chart reading effectively, their ability to perform comparative and quantitative reasoning from more complex visual structures remains limited.

Compared to Task 1, performance shows a noticeable decline, indicating that reasoning over chart data is more challenging than interpreting visual encodings. Gemini achieved the highest accuracy (74.0%), followed by ChatGPT (66.7%), Claude (56.9%), and Mistral (51.3%).

Table 5.18: Performance of LLMs on Task 2 (percentage of correct, partially correct, and incorrect responses).

LLMs	% Correct	% Partially Correct	% Incorrect
ChatGPT	66.70%	5.70%	27.60%
Gemini	74.00%	6.40%	19.60%
Claude	56.90%	9.00%	34.10%
Mistral	51.30%	6.00%	42.70%

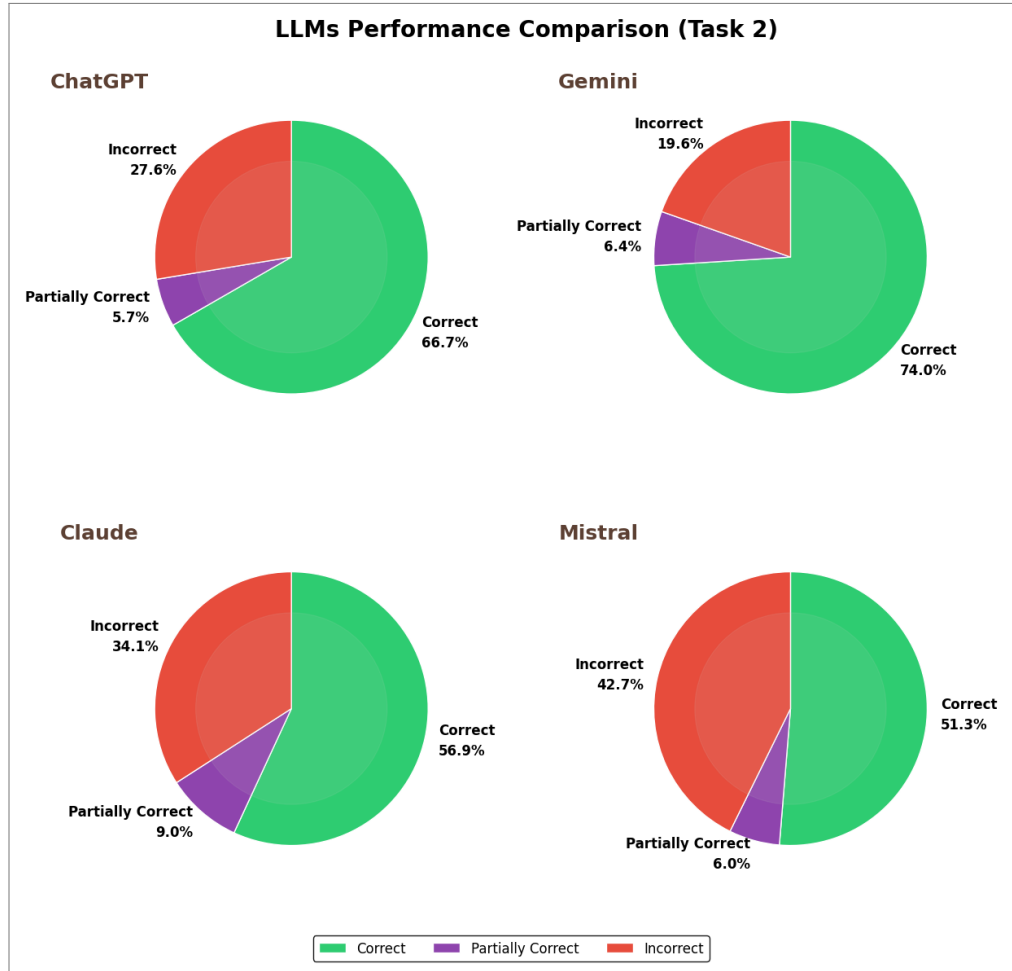


Figure 5.9: Visualisation of Task 2 performance of LLMs (percentage of correct, partially correct, and incorrect responses)

5.3.3 Task 3: Detection of Misleading Charts

In Task 3, LLMs generally performed well at identifying obvious and widely recognized visualization flaws, such as the use of 3D distortions, missing legends, or overly cluttered layouts. However, they frequently failed to detect more subtle perceptual manipulations. Commonly missed issues included incorrect proportional scaling, misleading normalization, and violations of statistical logic, such as percentages exceeding 100 percentage. Additionally, models often failed to identify inappropriate chart types, particularly when charts appeared visually plausible despite being conceptually incorrect. As observed in previous tasks, performance was stronger for simple and familiar chart types and deteriorated noticeably for more advanced visualizations. Overall,

errors related to visual perception and quantitative reasoning rather than surface-level design flaws were the most prevalent.

In contrast to Task 2, performance improves again, with Gemini (79.9%), Claude (79.0%), and ChatGPT (77.4%) achieving relatively high accuracy. Mistral, however, shows significantly lower performance (61.2%), indicating difficulty in this task.

Table 5.19: Performance of LLMs on Task 3 (percentage of correct, partially correct, and incorrect responses).

LLMs	% Correct	% Partially Correct	% Incorrect
ChatGPT	77.40%	4.30%	18.30%
Gemini	79.90%	3.40%	16.70%
Claude	79.00%	3.20%	17.80%
Mistral	61.20%	2.30%	36.50%

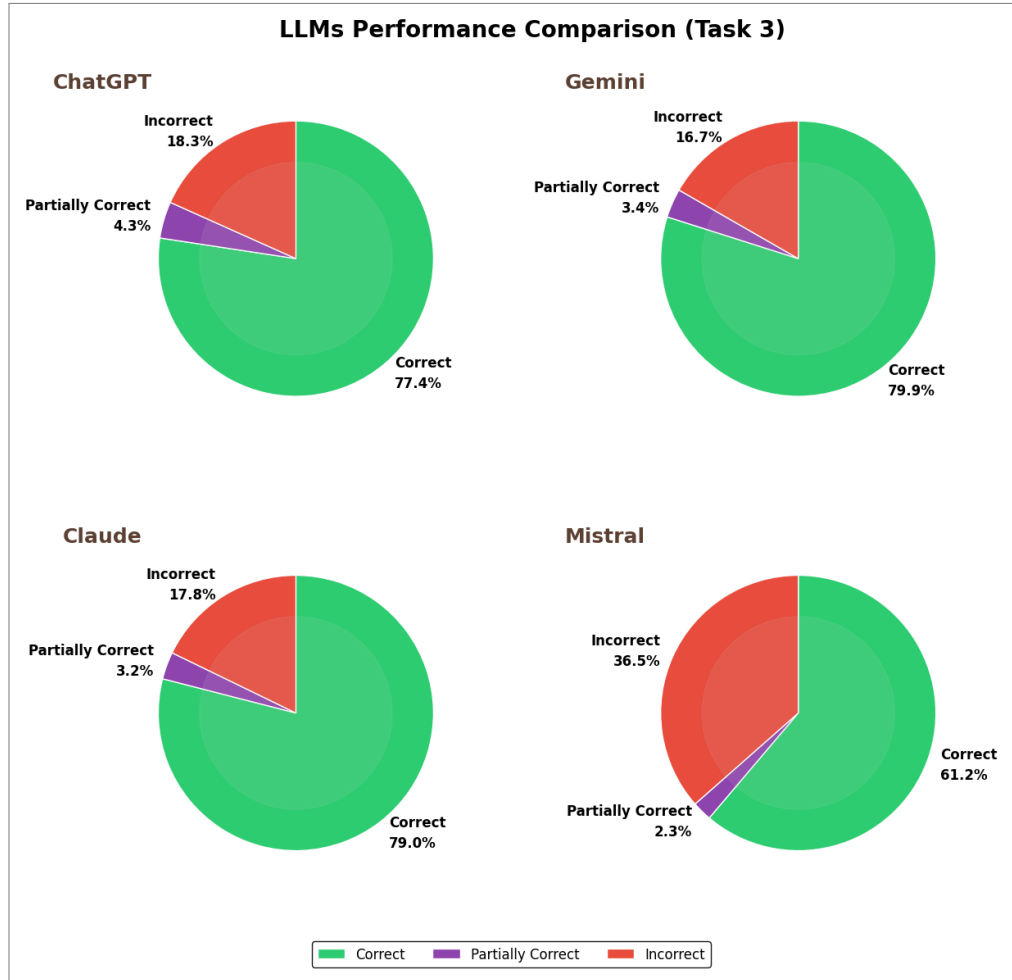


Figure 5.10: Visualisation of Task 3 performance of LLMs (percentage of correct, partially correct, and incorrect responses)

5.4 Summary of Results Across Tasks

The evaluation of LLMs across the three chart-related tasks reveals a clear dependence on chart complexity. Common charts, such as bar, line, and pie charts, were handled accurately and consistently, with correct data representation, labeling, and visual encoding. In contrast, Advanced charts including radar, network, and geospatial visualizations exposed notable limitations, with frequent structural errors, incomplete multi-dimensional representations, and misinterpretation of visual encodings. Additionally, while LLMs perform reliably on simple, single-variable charts, their accuracy and consistency decrease with increasing chart complexity and dimensionality. These

findings underscore the need for targeted improvements to enhance LLM reasoning and reliability in advanced visualization tasks.

The overall evaluation results demonstrate a clear relationship between task complexity and model performance. LLMs perform best in tasks involving direct visual interpretation, as seen in Task 1, where accuracy is consistently high. Performance declines in Task 2, where analytical reasoning and numerical precision are required, and partially improves in Task 3, where models rely on identifying visual inconsistencies.

Across all tasks, Gemini achieves the highest overall performance, followed closely by ChatGPT, while Claude demonstrates moderate performance with a higher tendency toward partially correct responses. Mistral consistently records the lowest accuracy, particularly in tasks requiring reasoning and critical evaluation.

These results indicate that LLMs are more reliable in handling perceptual and pattern recognition tasks than in performing precise analytical reasoning or evaluating complex visualizations.

ChatGPT demonstrates strong and consistent performance across all tasks, with particularly high accuracy in visual encoding interpretation and stable performance in analytical tasks. Its primary strength lies in its balanced capability across different task types. However, it still exhibits limitations in tasks requiring precise numerical reasoning and in identifying subtle misleading elements.

Gemini achieves the highest overall performance, particularly excelling in analytical reasoning and misleading chart detection. Its strength lies in its ability to identify patterns and extract relevant information accurately. Nevertheless, occasional partial correctness suggests minor inconsistencies in precise interpretation.

Claude demonstrates moderate performance and shows a higher proportion of partially correct responses. This indicates that while the model often captures the general intent or trend, it lacks precision in its final answers. Its performance is more sensitive to increasing task complexity, particularly in reasoning-intensive scenarios.

Mistral consistently records the lowest performance across all tasks. While it performs reasonably well in simpler visual encoding tasks, it struggles significantly with analyt-

ical reasoning and critical evaluation. Its higher error rates suggest limitations in both numerical reasoning and understanding of visualization principles.

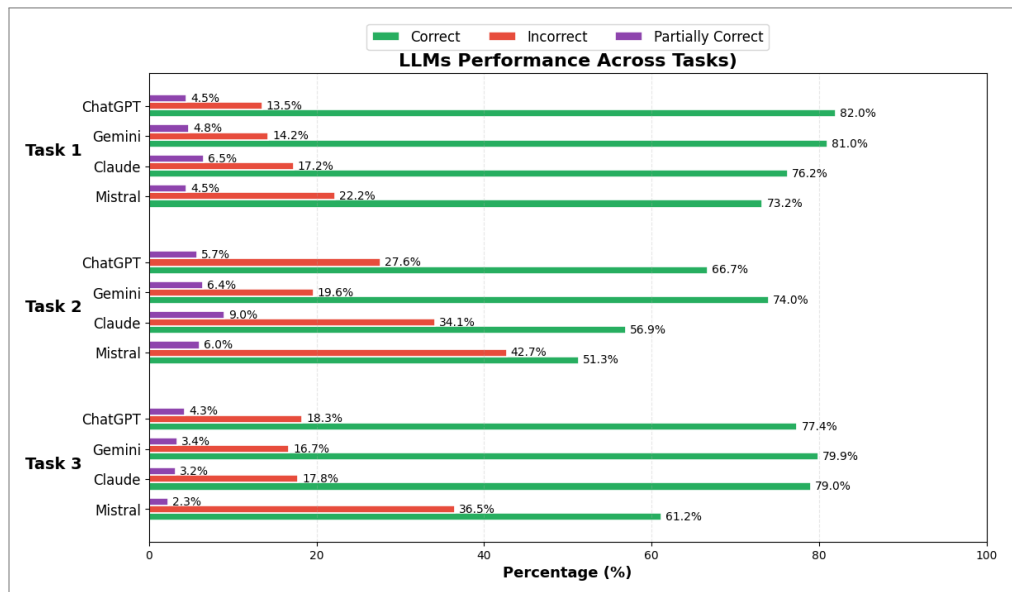


Figure 5.11: Performance of LLMs across Task 1, 2 and 3 (percentage of correct, partially correct, and incorrect responses).

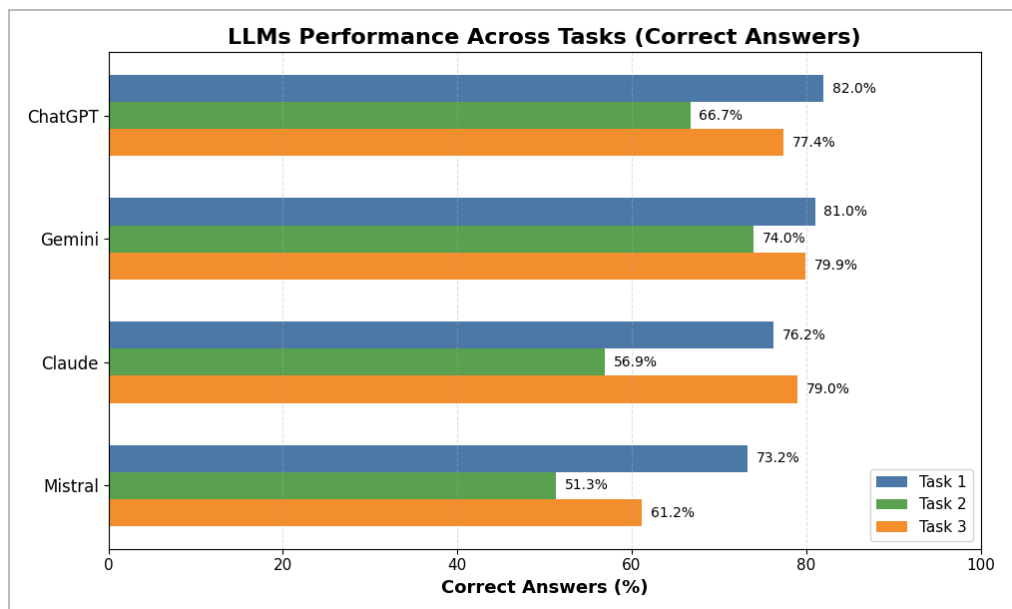


Figure 5.12: Performance of LLMs across Task 1, 2 and 3 for correct responses.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project presented the execution of a structured evaluation framework for assessing the baseline performance of multiple Large Language Models (LLMs) on chart-related tasks. The evaluation covered key areas including visual encoding interpretation, chart-based question answering, and detection of misleading visualizations.

The results indicate that while LLMs demonstrate promising capabilities in interpreting and reasoning, their performance is not consistently reliable across all tasks. Common error patterns were observed, particularly in tasks requiring precise visual interpretation and critical evaluation of misleading charts.

Overall, the findings provide a comprehensive understanding of the strengths and limitations of contemporary LLMs in visualization-related tasks, highlighting both their potential and areas requiring improvement.

6.2 Future Work

Building on the findings of this study, several directions for future work can be explored. One key extension involves the application of prompt engineering techniques to improve LLMs performance. By designing more structured and optimized prompts, it is possible

to guide models toward generating more accurate, consistent, and reliable outputs for chart-related tasks.

Additionally, future work can expand the scope of evaluation by incorporating a wider range of chart types and more diverse task categories, enabling a more comprehensive assessment of LLMs capabilities. Increasing the number of test cases and introducing more complex or real-world datasets can further strengthen the robustness of the evaluation.

Further improvements may also include refining the evaluation metrics and scoring framework to capture more nuanced aspects of model performance, such as reasoning quality and explanation clarity. These enhancements would contribute to a deeper and more detailed understanding of LLMs behavior in visualization contexts.

6.3 Final Remarks

This project highlights the growing role of LLMs in supporting data visualization and analysis, while emphasizing the importance of careful evaluation when applying these models in practice. The insights gained from this study establish a strong foundation for future improvements, particularly in enhancing model reliability and reducing errors in chart interpretation and analysis.

By combining systematic evaluation with targeted improvements such as prompt engineering, LLMs have the potential to become more effective and trustworthy tools in visualization-driven applications.

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